

Exponential and logarithmic functions



Outline

Exponential functions

Logarithmic functions

Change of base

Exponential functions



Introduction: Exponential functions

- ▶ In this section we study the **exponential functions** $x \mapsto b^x$, in which the variable appears in the exponent
- ▶ We will see that they are always **positive** and that their **monotonicity** depends on the base b
- ▶ Exponential functions describe growth and decay (compound interest, populations, radioactive decay)

Exponential functions

Definition 1: exponential functions

Given $b \in \mathbb{R}_+ \setminus \{1\}$, the function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto b^x$$

is called the **exponential function** with base b .

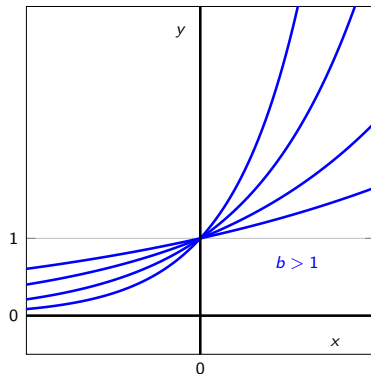
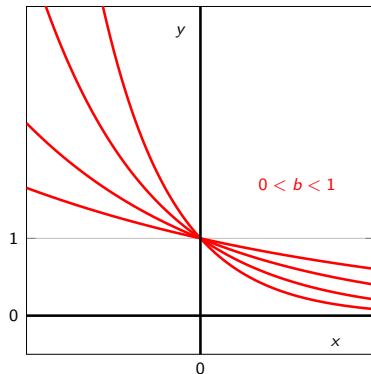
- ▶ Regarding positivity and monotonicity, we have:

$$b^x > 0, \quad \forall x \in \mathbb{R}$$

and

- ▶ if $0 < b < 1$ then f is decreasing on $\mathbb{R}$
- ▶ if $b > 1$ then f is increasing on $\mathbb{R}$

Exponential functions: graphs



Summary

Exponential functions:

- ▶ $f(x) = b^x$ with $b > 0$, $b \neq 1$, defined on the whole of $\mathbb{R}$
- ▶ $b^x > 0$ for every $x \in \mathbb{R}$
- ▶ If $0 < b < 1$ the function is **decreasing**, if $b > 1$ it is **increasing**

Logarithmic functions



Introduction: Logarithmic functions

- ▶ In this section we study the **logarithmic functions**
 $x \mapsto \log_a x$, defined for $x > 0$
- ▶ We analyze their **sign** and **monotonicity**, which depend on the base a
- ▶ Logarithms allow us to solve exponential equations and to handle quantities that vary over many orders of magnitude

Logarithmic functions

Definition 2: logarithmic functions

Given $a \in \mathbb{R}_{>0} \setminus \{1\}$, the function:

$$f : (0, +\infty) \rightarrow \mathbb{R}, \quad f : x \mapsto \log_a x$$

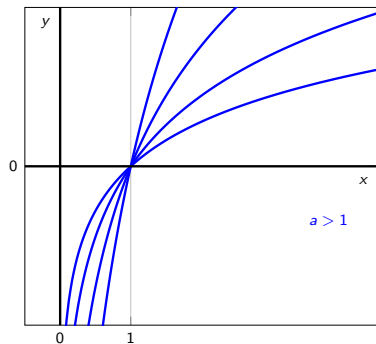
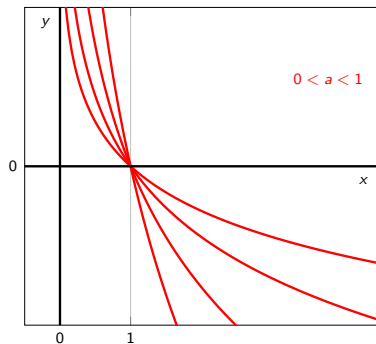
is called the **logarithmic function** with base a .

Sign and monotonicity

- ▶ Regarding positivity and monotonicity, we have:
 - ▶ if $0 < a < 1$: $\log_a x > 0$, $\forall x \in (0, 1)$,
 $\log_a x < 0$, $\forall x \in (1, +\infty)$ and f is decreasing on $\mathbb{R}_{>0}$
 - ▶ if $a > 1$: $\log_a x < 0$, $\forall x \in (0, 1)$, $\log_a x > 0$, $\forall x \in (1, +\infty)$
and f is increasing on $\mathbb{R}_{>0}$
- ▶ Moreover, we have:

$$\log_a x = 0 \iff x = 1, \quad \forall a \in \mathbb{R}_{>0} \setminus \{1\}$$

Logarithmic functions: graphs



Summary

Logarithmic functions:

- ▶ $f(x) = \log_a x$ with $a > 0$, $a \neq 1$, defined on $(0, +\infty)$
- ▶ $\log_a x = 0 \iff x = 1$
- ▶ If $0 < a < 1$ the function is **decreasing**, if $a > 1$ it is **increasing**; the sign changes when crossing $x = 1$

Change of base

Introduction: Change of base

- ▶ In this section we see how to move from one base to another for exponentials and logarithms
- ▶ Every exponential and every logarithm can be rewritten in another base through a **multiplicative constant** λ
- ▶ For this reason it is enough to know one base well, typically Euler's number e

Change of base

Given any base $c \in \mathbb{R}_{>0} \setminus \{1\}$, all exponential and logarithmic functions can be rewritten in terms of another base $d \in \mathbb{R}_{>0} \setminus \{1\}$ as follows:

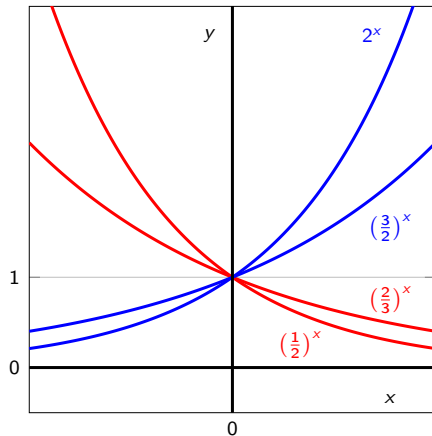
► $c^x = d^{\log_d c^x} = d^{x \log_d c}$, hence with $\lambda = \log_d c$ we get
 $c^x = d^{\lambda x}$

► $\log_c x = \frac{\log_d x}{\log_d c}$, hence with $\lambda = \frac{1}{\log_d c}$ we get
 $\log_c x = \lambda \log_d x$

The logarithmic function with base e is also written $\log x$ or $\ln x$, while the one with base 2 is also written $\lg x$.

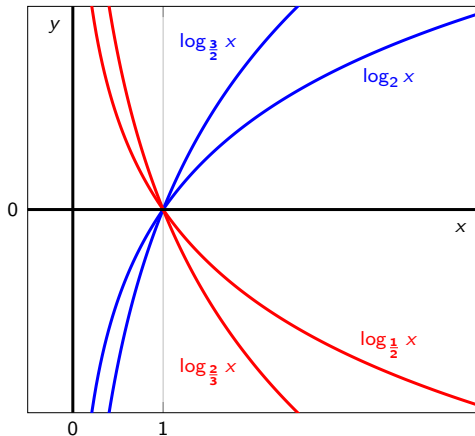
Example (1/2): graphs of exponential and logarithmic functions

Exponential functions:



Example (2/2): graphs of exponential and logarithmic functions

Logarithmic functions:



Summary

Change of base:

- ▶ $c^x = d^{\lambda x}$ with $\lambda = \log_d c$
- ▶ $\log_c x = \lambda \log_d x$ with $\lambda = \frac{1}{\log_d c}$
- ▶ Notation: $\log x = \ln x$ (base e), $\lg x$ (base 2)

Final summary

- ▶ **Exponentials** b^x : defined on $\mathbb{R}$, always positive, increasing if $b > 1$ and decreasing if $0 < b < 1$
- ▶ **Logarithms** $\log_a x$: defined on $(0, +\infty)$, zero at $x = 1$, increasing if $a > 1$ and decreasing if $0 < a < 1$
- ▶ With the **change of base**, every exponential and every logarithm can be rewritten in a fixed base up to a multiplicative constant